

Symmetric-tensor leverage scores give a universal answer to Problem 1 on $\mathcal{Z}_p$ -norms

Candidate full solution to Problem 1 in arXiv:1711.02214

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Status. Candidate full solution to the first formulation of Problem 1, pending expert review. The stronger p -moment formulation is also proved for every vector with regular one-dimensional moment growth, including centered log-concave vectors. Problem 2 is proved for $q \leq p$; its remaining high-moment range is not claimed.

Source. R. Latała, *On $\mathcal{Z}_p$ -norms of random vectors*, arXiv:1711.02214, Problems 1 and 2 on printed pages 2–3 [1].

1 The questions and definitions

Let X be an n -dimensional random vector and $p \geq 2$. The source defines

$$\|t\|_{\mathcal{M}_p(X)} = (\mathbb{E}|\langle t, X \rangle|^p)^{1/p}, \quad \|x\|_{\mathcal{Z}_p(X)} = \sup_{\|t\|_{\mathcal{M}_p(X)} \leq 1} |\langle t, x \rangle|.$$

Thus the $\mathcal{Z}_p(X)$ -norm is the dual of the marginal-moment norm.

Transcription of Problem 1. Is it true that, for at least a large class of centered n -dimensional random vectors X ,

$$(\mathbb{E}\|X\|_{\mathcal{Z}_p(X)}^2)^{1/2} \leq C \sqrt{\frac{n+p}{p}}, \quad p \geq 2,$$

or perhaps even with the outer exponent p in place of 2?

Problem 2 asks, for centered log-concave X , $2 \leq p \leq n$, and $1 \leq q \leq \sqrt{pn}$, whether

$$(\mathbb{E}\|X\|_{\mathcal{Z}_p(X)}^q)^{1/q} \leq C \sqrt{\frac{n}{p}}.$$

$$(\mathbb{E}\|AX\|_{\mathcal{Z}_p(X)}^p)^{1/p} = \|U_1\|_p \|\mathbf{R}\|_p (\mathbb{E}\|A\|_F^2)^{1/2} = \|U_1\|_p \sim \sqrt{\frac{n+p}{p}}. \quad (1)$$

This motivates the following problem.

Problem 1. Is it true that for (at least a large class of) centered n -dimensional random vectors X ,

$$\left(\mathbb{E}\|X\|_{\mathcal{Z}_p(X)}^2\right)^{1/2} \leq C\sqrt{\frac{n+p}{p}} \quad \text{for } p \geq 2,$$

or maybe even

$$\left(\mathbb{E}\|X\|_{\mathcal{Z}_p(X)}^p\right)^{1/p} \leq C\sqrt{\frac{n+p}{p}} \quad \text{for } p \geq 2?$$

Notice that the problem is linearly-invariant, since

$$\|AX\|_{\mathcal{Z}_p(AX)} = \|X\|_{\mathcal{Z}_p(X)} \quad \text{for any } A \in \text{GL}(n). \quad (2)$$

For any centered random vector X with nondegenerate covariance matrix, random vector $Y = \text{Cov}(X)^{-1/2}X$ is isotropic (i.e. centered with identity covariance matrix). We have $\mathcal{M}_2(Y) = \mathcal{Z}_2(Y) = B_2^n$, hence

$$\mathbb{E}\|X\|_{\mathcal{Z}_2(X)}^2 = \mathbb{E}\|Y\|_{\mathcal{Z}_2(Y)}^2 = \mathbb{E}|Y|^2 = n.$$

Next remark shows that the answer to our problem is positive in the case $p \geq n$.

Remark 1. For $p \geq n$ and any n -dimensional random vector X we have $(\mathbb{E}\|X\|_{\mathcal{Z}_p(X)}^p)^{1/p} \leq$

Figure 1: Problem 1 in the archived source PDF, printed page 2.

2 Proof Intuition

When the exponent is even, say $2k$, the scalar $\langle t, x \rangle^k$ is a linear functional after lifting x to its symmetric tensor $x^{\otimes k}$. The denominator in the definition of $\mathcal{Z}_{2k}$ becomes the quadratic form of the lifted second-moment matrix

$$A = \mathbb{E}[X^{\otimes k}(X^{\otimes k})^*].$$

The original supremum only ranges over rank-one tensors $t^{\otimes k}$. Enlarging it to the whole symmetric tensor space turns the bound into an ordinary leverage score. Its expectation is exactly $\text{rank } A$, which is at most $\dim \text{Sym}^k(\mathbb{R}^n) = \binom{n+k-1}{k}$.

This argument does not use signs, independence, centering, or log-concavity. Rounding p down to an even exponent gives the requested outer second moment for arbitrary p . If marginal moments grow regularly, rounding upward gives the stronger outer p -moment estimate.

3 The tensor leverage estimate

Equip $H_k = \text{Sym}^k(\mathbb{R}^n)$ with the Hilbert structure inherited from the tensor product, so that

$$\langle t^{\otimes k}, x^{\otimes k} \rangle = \langle t, x \rangle^k, \quad \dim H_k = \binom{n+k-1}{k}.$$

Lemma 1 (even-moment master bound). *Let $k \geq 1$, and suppose X has a finite $2k$ th moment. Then*

$$\mathbb{E}\|X\|_{\mathcal{Z}_{2k}(X)}^{2k} \leq \text{rank}(\mathbb{E}[X^{\otimes k}(X^{\otimes k})^*]) \leq \binom{n+k-1}{k}. \quad (1)$$

No symmetry or centering assumption is needed.

Proof. Write $\Phi(x) = x^{\otimes k}$ and $A = \mathbb{E}[\Phi(X)\Phi(X)^*]$ on H_k . First note that $\Phi(X) \in \text{ran } A$ almost surely. Indeed, for $v \in \ker A$,

$$0 = \langle v, Av \rangle = \mathbb{E}|\langle v, \Phi(X) \rangle|^2.$$

Applying this to a finite orthonormal basis of $\ker A$ proves the assertion simultaneously outside one null set.

For every x in that full-measure set, homogeneity and the definition of the dual norm give

$$\begin{aligned}
\|x\|_{\mathcal{Z}_{2k}(X)}^k &= \sup_{t \neq 0} \frac{|\langle t, x \rangle|^k}{(\mathbb{E}|\langle t, X \rangle|^{2k})^{1/2}} \\
&= \sup_{t \neq 0} \frac{|\langle t^{\otimes k}, \Phi(x) \rangle|}{\langle t^{\otimes k}, A t^{\otimes k} \rangle^{1/2}} \\
&\leq \sup_{v \in H_k: \langle v, Av \rangle > 0} \frac{|\langle v, \Phi(x) \rangle|}{\langle v, Av \rangle^{1/2}} = \langle \Phi(x), A^\dagger \Phi(x) \rangle^{1/2},
\end{aligned} \tag{2}$$

where $A^\dagger$ is the Moore–Penrose inverse. The last equality follows by diagonalizing A on its range; vectors in its kernel have zero numerator. Squaring (2), taking expectations, and using finite-dimensional trace cyclicity yields

$$\mathbb{E}\|X\|_{\mathcal{Z}_{2k}(X)}^{2k} \leq \text{tr}(A^\dagger \mathbb{E}[\Phi(X)\Phi(X)^*]) = \text{tr}(A^\dagger A) = \text{rank } A.$$

The dimension of H_k gives the final inequality. $\square$

4 Full answer to Problem 1

Theorem 1. *Let X be an n -dimensional random vector with finite p th moment.*

1. *For every real $p \geq 2$,*

$$(\mathbb{E}\|X\|_{\mathcal{Z}_p(X)}^2)^{1/2} \leq 2\sqrt{e} \sqrt{\frac{n+p}{p}}. \tag{3}$$

2. *If $p = 2k$ is even, then the stronger exact estimate*

$$(\mathbb{E}\|X\|_{\mathcal{Z}_p(X)}^p)^{1/p} \leq \binom{n+k-1}{k}^{1/(2k)} \tag{4}$$

holds.

3. *Suppose, more generally, that for some $\lambda \geq 1$ every marginal satisfies*

$$\|\langle t, X \rangle\|_s \leq \lambda \frac{s}{q} \|\langle t, X \rangle\|_q, \quad s \geq q \geq 2. \tag{5}$$

Then for every real $p \geq 2$,

$$(\mathbb{E}\|X\|_{\mathcal{Z}_p(X)}^p)^{1/p} \leq 2\sqrt{2e} \lambda \sqrt{\frac{n+p}{p}}. \tag{6}$$

Proof. For (3), set $k = \lfloor p/2 \rfloor$ and $r = 2k$. Since $r \leq p$, monotonicity of scalar moments gives $\|t\|_{\mathcal{M}_r(X)} \leq \|t\|_{\mathcal{M}_p(X)}$, and polarity therefore gives $\|x\|_{\mathcal{Z}_p(X)} \leq \|x\|_{\mathcal{Z}_r(X)}$. As $r \geq 2$, Lemma 1 implies

$$\|X\|_{\mathcal{Z}_p(X)} \leq \|X\|_{\mathcal{Z}_r(X)} \leq \binom{n+k-1}{k}^{1/(2k)}.$$

Now $k \geq p/4$ and $\binom{N}{k} \leq (eN/k)^k$, whence

$$\binom{n+k-1}{k}^{1/(2k)} \leq \sqrt{e \frac{n+k-1}{k}} \leq 2\sqrt{e} \sqrt{\frac{n+p}{p}}.$$

Part (4) is Lemma 1 with $p = 2k$.

For (6), take $k = \lfloor p/2 \rfloor$ and $r = 2k$, so $p \leq r \leq 2p$. By (5), $\|t\|_{\mathcal{M}_r(X)} \leq \lambda(r/p)\|t\|_{\mathcal{M}_p(X)}$. Polarity gives $\|x\|_{\mathcal{Z}_p(X)} \leq \lambda(r/p)\|x\|_{\mathcal{Z}_r(X)}$. Consequently, Lemma 1 and $k \geq p/2$ give

$$\begin{aligned}
\|X\|_{\mathcal{Z}_p(X)} &\leq \lambda \frac{r}{p} \|X\|_{\mathcal{Z}_r(X)} \\
&\leq 2\lambda \binom{n+k-1}{k}^{1/(2k)} \leq 2\sqrt{2e} \lambda \sqrt{\frac{n+p}{p}}.
\end{aligned}$$

$\square$

The first part is stronger than the first formulation of Problem 1: it holds for every X , without centering. The source records (5) with a universal λ for centered log-concave random vectors [1]; hence the stronger p -moment formulation is affirmative for that principal class.

5 Consequences and the remaining part of Problem 2

Corollary 1. *If X is centered and log-concave, $2 \leq p \leq n$, and $1 \leq q \leq p$, then*

$$(\mathbb{E}\|X\|_{\mathcal{Z}_p(X)}^q)^{1/q} \leq C\sqrt{\frac{n}{p}}.$$

Proof. Outer-moment monotonicity and (6) give the bound with $\sqrt{(n+p)/p} \leq \sqrt{2n/p}$. $\square$

This proves Problem 2 through $q = p$. For $q > p$, applying regular moment growth and the tensor lemma at an even exponent comparable to q gives only

$$(\mathbb{E}\|X\|_{\mathcal{Z}_p(X)}^q)^{1/q} \lesssim \lambda \frac{q}{p} \sqrt{\frac{n+q}{q}}. \quad (7)$$

At $q = \sqrt{pn}$, (7) is of order $(n/p)^{3/4}$ rather than the requested $(n/p)^{1/2}$. Eight focused upgrade routes are documented in the companion attempt note; the unresolved range is tied to the general weak/strong moment problem already identified by the source. No claim is made for $p < q \leq \sqrt{pn}$.

6 Verification, scope, and novelty

The script `code/verify_tensor_leverage.py` checks the tensor normalization on 80 weighted finite distributions in $\mathbb{R}^2$, for $1 \leq k \leq 5$. It includes singular lifted moment matrices, verifies $\mathbb{E}\langle \Phi, A^\dagger \Phi \rangle = \text{rank } A$, and compares a dense rank-one directional supremum with the full-tensor leverage bound. All checks pass. This is a normalization and singular-case check, not a substitute for the proof.

The theorem assumes the source’s “ n -dimensional” condition so that the marginal functional is a norm. Degenerate distributions are handled by restricting to their linear span, with n replaced by its dimension. The only remaining limitation relative to the two source problems is the high- q portion of Problem 2 stated above.

Novelty check. On 12 August 2026, the run indexes, arXiv/web search, and exact-title searches were queried with the source id and combinations of “ $\mathcal{Z}_p(X)$ norm”, “symmetric tensor”, “tensor leverage score”, “centroid body”, “total ℓ_p sensitivity”, and the binomial dimension $\binom{n+k-1}{k}$. Searches also inspected the later algorithmic literature on total ℓ_p sensitivity [2]. That literature recognizes the equivalent discrete sensitivity quantity, but the bounded inspection did not locate (1), the universal answer (3), or an explicit answer to Latała’s Problem 1. Novelty confidence is moderate, not definitive.

Human-review recommendation. Prioritize expert review. The critical checks are the enlargement from rank-one tensors to all of H_k , the almost-sure range statement when A is singular, and the polarity directions in the two rounding arguments. Each is isolated in Lemma 1 and Theorem 1.

References

- [1] R. Latała, *On $\mathcal{Z}_p$ -norms of random vectors*, Zapiski Nauchnykh Seminarov POMI **457** (2017), 211–225; arXiv:1711.02214.
- [2] S. Padmanabhan, D. P. Woodruff, and Q. Zhang, *Computing approximate ℓ_p sensitivities*, arXiv:2311.04158 (2023).